

Room-Scanner: submission summary

Phone capture (photos, video or LiDAR) in → a dimensioned floor plan out (plan.json to a schema + plan.svg), with walls, ceiling height, floor area, doors/windows, room adjacency, damage regions, concealed-damage flags and scope line items, and a 90% interval on every number. One command per capture: `roomscan run <capture> -o out/`. Built in 48 h on an 8 GB M1 MacBook Air; captures from an iPhone 14 Plus (no LiDAR on hand). Pipeline v0.9.0.

What works (measured)

	Result	Evidence
Photo tier, one room, protocol capture	walls -1.6% / +4.5%, area +0.7%, ceiling -2.0 cm vs tape	BENCHMARK §1
Interval calibration	90% intervals hold in 8–9 of 10 captures of the taped room	E26, BENCHMARK §2b
Repeatability	3 live runs identical to 4 decimals (photos and video)	BENCHMARK §2
Fix loop	declared worst gate (video short side -35.8%), fixed to -4.0%, predictions recorded	fixloop/
LiDAR	drift on revisits 19.7 → 3.5 cm; rooms split with adjacency	E7, E19
Robustness	all 25 captures on disk complete; off-protocol captures run with warnings; clean-machine install ≈ 10 min	BENCHMARK §5b–5c
Damage	local detector (OWLv2, no API key); 0 false detections on 17 clean photos (in-sample)	E24

What does not work (and why)

- **Stitching rooms into one plan from photos** is limited by evidence: only rooms that some photo sees *into* can be placed. Our house photos rarely look between rooms, so 3 of 6 rooms link with the default matcher (5 of 7 with MAST3R on a GPU, unverified layout). Unlinked rooms are measured and drawn apart, clearly marked, never guessed. The capture protocol's look-through photo per doorway is the fix (E14, E28–E32).
- **Video across several rooms** is not reliable: COLMAP splits a handheld walk into many small pieces whose joins disagree, so walls double and rooms merge (E21–E23, E34). Single-room video: walls -4% / -7% (fails the ±3% gate).
- **Not done:** head-to-head vs magicplan and LiDAR accuracy (no Pro iPhone); damage recall (no staged damage); only one room has tape, so all accuracy claims rest on it.

Where to look

- README.md: setup and quick start · docs/WALKIN.md: runbook
- docs/TECH_REPORT.md: method, uncertainty, failure modes · docs/BENCHMARK.md: all numbers
- docs/EXPERIMENTS.md: E1–E34, every experiment incl. negative results · docs/CHANGELOG.md: versioned history
- docs/COMPLIANCE.md: requirement → file · fixloop/: the fix-loop bundle · docs/CAPTURE_PROTOCOL.md

Room-Scanner: technical report

Pipeline 0.9.0 · M1 MacBook Air 8 GB (lite profile) · iPhone 14 Plus (no LiDAR) · 2026-10-03. Companion documents: docs/BENCHMARK.md (numbers), docs/EXPERIMENTS.md (E1–E21, how each was found), docs/CHANGELOG.md (versioned history), fixloop/ (declared fix), docs/COMPLIANCE.md (requirement → file).

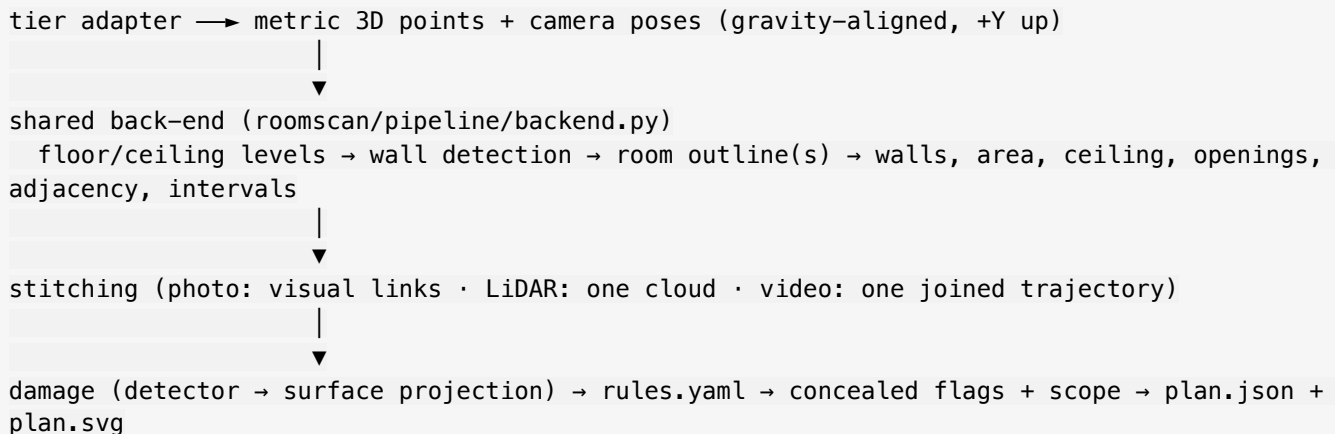
1. Problem and what was built

A phone capture in one of three tiers (photos, video, LiDAR) goes in. Out comes `plan.json` (validated against `schema/plan.schema.json`) and `plan.svg`. They contain per-room walls, ceiling height, floor area and openings, the rooms stitched into one plan with adjacency, damage regions on surfaces, concealed-damage flags and scope line items. Every number carries a 90% interval. One command per capture: `roomscan run <capture> -o out/` (the tier is detected from the input).

Capture route: Route 2, the stock protocol (`docs/CAPTURE_PROTOCOL.md`). No custom app: a 0.5× overlapping ring of landscape stills per room, a slow 0.5× video walk, or a Stray Scanner LiDAR scan. Plus one photo per doorway looking into the next room, which is what lets photo rooms be linked (E14).

Constraint that shaped everything: an 8 GB laptop with a shared GPU. It crashed once at 4.46 GB of GPU memory (E8). Since then the GPU is capped, one model is loaded at a time, and a watchdog (`scripts/run_guarded.sh`) kills a run before the machine swaps. Stronger machines get bigger models through hardware profiles (`configs/profiles/`).

2. Architecture



2.1 Tier adapters

- **LiDAR** (`roomscan/io/stray.py`, `pipeline/lidar.py`). Stray Scanner depth (256×192, confidence ≥ 2) fused with ARKit poses. ARKit's poses are camera-to-world in the OpenCV convention. Getting that convention wrong put the floor at camera height until it was fixed.
- **Photos** (`pipeline/photos.py`, `recon/learned.py`). Depth Anything 3 Base gives multi-view depth and relative poses. DA3Metric-Large fixes metric scale with the photo's **EXIF focal length**. E1 showed the predicted focal length was off by up to +24% and scale errors reached +18%. The true focal length gives −1 to −9% with the small model, as good as the 1.4 B Giant model, which is why the 8 GB profile is viable. Views are processed in overlapping windows chained by Sim(3) transforms on shared views, because memory grows steeply with view count (E1).

- **Video** (`pipeline/video.py`, `recon/sfm.py`, `recon/bridge.py`). ffmpeg decodes frames sequentially (HEVC seeking returned wrong frames). COLMAP (`pycolmap`) tracks cameras. It runs in a child process because its OpenMP runtime aborts alongside torch's. It is seeded and single-threaded so results repeat, keeping the best of 4 seeds. COLMAP splits a handheld walk into pieces at fast turns. Each piece gets its own metric scale from DA3Metric depth against its sparse points, and pieces are joined by DA3 bridges at their nearest frames (the fix loop, §5). Keyframe metric depth placed at the joined poses gives the dense cloud.

2.2 Geometry back-end (shared by all tiers)

- **Floor and ceiling:** height histogram of the cloud. The floor is the *lowest* flat level with $\geq 40\%$ of the densest level's support, and is accepted only if the camera is a plausible 1.0–1.95 m above it. Otherwise no ceiling is reported, rather than a wrong one (E12). Before this rule, desks and beds became the "floor" and the ceiling was wrong by up to 1.5 m. The ceiling is the median of the top layer inside the room outline.
- **Walls:** points covering $\geq 35\%$ of the 0.2–1.8 m height band in a top-down grid, rotated to the dominant wall direction.
- **Room outline:** LiDAR and video carve free space with camera→point rays (thousands of viewpoints) and split rooms with a watershed. Photos can't carve with only 4–8 viewpoints (E6: 28% of a room even with perfect depth), so the outline is the region enclosed by observed walls. A Manhattan-box alternative was tested and rejected (E13).
- **Openings** (`openings/detect.py`): 5 cm slots along each wall. A *see-through* opening is a run with little wall surface where points exist beyond the wall line. A *gap* opening (photos only, confidence 0.3) is a run with no wall surface inside a camera's view while the rest of the wall is covered. Tolerances depend on the tier: LiDAR walls are ± 1 cm thick, learned-depth walls ± 20 cm (E15).
- **Adjacency:** rooms from one cloud are adjacent through a door detected from both sides (within 0.6 m), or through an open passage where the carved floor directly between them connects them (E19).

2.3 Stitching

- **LiDAR** is one cloud, so stitching is free, but ARKit drifts (19.7 cm where the 215 s walk revisits places). **Drift correction** (`stitch/drift.py`, E7): the walk is cut into 4 s fragments. Revisited fragments are matched as top-down wall images (yaw search $\pm 3^\circ$, phase correlation, accepted only if wall overlap improves and the shift is under 30 cm). A small linear pose graph is then solved. Revisit misalignment fell 19.7 → 3.5 cm and walls got sharper. On the short walk no revisit passed the test, and nothing moved, by design. Open3D ICP was tried first and diverged (40–90° rotations).
- **Photos** (`stitch/links.py`, `stitch/photo_graph.py`, E14, E17, E18). Each room is reconstructed in its own frame. DISK + LightGlue matches photos across rooms (SIFT fails on 0.5× ring photos: 18 median matches even within a room). A link gives a relative pose by PnP on the matched points' metric depth. It is accepted only if both directions agree (yaw within 10° , tilt within 15°), because match counts on the Mac GPU are not bit-reproducible, so a fixed inlier threshold flickered. Yaw is snapped to the wall grid. Then the two rooms' facing walls are snapped to one wall thickness apart and their doors aligned (PnP gets direction right but distance poorly). Rooms with no link are drawn to the side with a warning, never guessed into place.

2.4 Damage, concealed flags, scope

- **Detector** (pluggable). Default: OWLv2 run locally (`damage/detect_local.py`, no key; defect queries plus negative queries for clean surfaces, 0 false detections on 17 clean photos, E24). Optional: one Claude `claude-opus-5-5` vision request per room (`damage/detect_claude.py`) with up to 8 views, structured JSON-schema output (class + box per image on a 0–1000 grid), server-side fallback

enabled. Responses are cached by content hash in `cache/vlm/` and committed, so damage replays bit-identically on another machine (the brief accepts cached model outputs; `damage.cache=live` forces a fresh call).

- **Projection** (`damage/project.py`): the box's pixels are back-projected with the view's metric depth and gravity-aligned pose. The surface is decided by height: floor (< 0.15 m), ceiling (within 0.15 m of it), else the nearest wall within 0.5 m. The region is the 5–95% extent in the surface's own (along, up) metres. The same stain seen in several views is merged (same surface and class, $\text{IoU} \geq 0.2$).
- **Rules** (`scope/rules.yaml`, `scope/engine.py`): explicit, readable rules. For example, R-WATER-BASE: a water stain whose bottom is < 0.30 m above the floor flags possible wicking into the wall cavity and adds a flood cut and lower drywall replacement. R-CEIL-STAIN, R-MOLD-ANY and R-CRACK-LONG (≥ 0.6 m) work the same way. Every flag and line item records the rule that produced it and the region ids that triggered it. Quantities come from wall and room dimensions, with intervals.

2.5 Uncertainty

Each tier has an error model of absolute (m) + relative (scale) terms. LiDAR: 1 cm + 0. Photos: 3 cm + 5% for a protocol capture, 15% for an off-protocol room (1× lens, portrait, < 4 photos), 25% without an EXIF focal length (E26: on 10 captures of the taped study, off-protocol sets were 9–21% off while claiming $\pm 9\%$). Video: 2 cm + a **measured** scale term, $\sqrt{(4\%^2 + \text{per-piece scale standard error}^2)}$, where each COLMAP piece's scale is a median of n keyframe depth ratios (E21). A fixed 4% was overconfident: the per-keyframe spread is 20–29%, giving 9.6% on the study video and 14.9% on the house walk. A wall's sigma combines both fitted neighbouring wall lines (corners are intersections) with the tier terms. Area combines independent wall-position noise with the common scale factor in quadrature (E26: an earlier version counted the scale term twice). Ceilings combine the top-layer spread, floor-level spread and tier terms. Intervals are $\pm 1.645\sigma$.

Coverage on every capture of the taped study (E26): photos 8/10 short side, 9/10 long side, 9/10 area, 2/2 ceiling; video 1/1 each. One room only and the off-protocol term is fitted on the same captures, so this supports the model rather than calibrating it.

2.6 Engineering for the walk-in test

- Checkpoints keyed by a hash of inputs + settings + code version (`recon/checkpoint.py`) mean an interrupted run resumes, and a re-run replays in ~ 7 s. `ROOMSCAN_NO_CACHE=1` reruns everything live.
- Models load offline-first. A mid-run Hugging Face disconnect killed a job once, and a stalled request hung the full-house run.
- Layered config with unknown keys rejected; `config.resolved.yaml` with a digest is written next to every output.
- A local website (`roomscan serve`) runs one guarded job at a time and compares results against typed-in tape.
- A memory watchdog (`scripts/run_guarded.sh`) measures macOS `phys_footprint`, which includes GPU and swap. It used to read RSS, which missed both: a video run thrashed 2.1 GB of swap unseen.
- A robustness sweep runs every capture on disk (25: photo variants, LiDAR, videos). It found that any fresh multi-room photo capture crashed at room 2, because the damage detector's GPU cache starved the next room's reconstruction. That is fixed: damage now runs after all geometry. The sweep is 25/25.

3. Results (details: `docs/BENCHMARK.md`)

Tier	Taped room result	Gate
Photos (study, 4 stills)	walls -1.6% / $+4.5\%$, area $+0.7\%$, ceiling -2.0 cm (interval holds)	walls ✓ area ✓ ceiling ✗
Video (study, off-protocol 1× walk)	walls -4.0% / -6.9% , area -10.8% (interval holds), no ceiling	✗ ($\pm 3\%$)

Tier	Taped room result	Gate
LiDAR	no taped LiDAR capture	n/a

- **Repeatability:** three live runs each of the study photos and the study video are identical to 4 decimals.
- **Whole-house video walk (E21):** 62% of frames tracked, 3 merged rooms instead of 6. Not usable, and reported as such: the intervals are wide.
- **LiDAR self-consistency:** 5 rooms with adjacency on the 215 s walk, drift 19.7 → 3.5 cm.
- **Connected plans:** photos connect 3 of 6 house rooms (the three with a look-through photo); LiDAR connects 5/5.
- **Runtime (live):** 15–28 s for one photo room, 136 s for the 6-room house (cross-room matching included), ~5 min for a 64 s video, 3.5 min for a 215 s LiDAR walk. Peak memory, including GPU: 4.2 GB for the live 6-room photo run, under the 6 GB cap (BENCHMARK §5).

4. What did not work (and what that taught)

- **Ray carving for photos** gave rooms 43–69% too small (E6). It needs thousands of viewpoints.
- **Proxy "photos" from LiDAR video frames** (E1, E6) were 1× portrait shots facing away from each other. They broke pose estimation in every model. The protocol now requires an overlapping 0.5× ring.
- **A fixed match threshold for photo links** was not reproducible on the Mac GPU (47 vs 42 matches for one pair). Two-way pose agreement replaced it (E18).
- **LiDAR open-passage adjacency** by mask distance linked rooms through walls. The connected-carved-floor rule fixed most cases; one probably-false link remains and is marked unverified (E19).
- **The first fix-loop implementation** joined COLMAP pieces in time order and dropped the largest piece (v1). The shipped version anchors on it. Both are kept on record.
- **Registering more video frames** two ways: looser COLMAP thresholds (E22) and DISK+LightGlue matches inside COLMAP (E23). Both registered far more frames (79%, 94%) and both made the room worse (–16% and +75% walls), because the extra poses were wrong. Fewer correct poses beat more wrong ones.
- **One multi-view model on all photos at once** (E29): DA3 on all 19 house photos stacked rooms on top of each other; no model can place rooms that no photo sees together.
- **Stitching with a stronger matcher** (E31–E32): MAST3R on a Kaggle GPU found far more cross-room matches and, with group merging and an overlap rule, linked 5 of 7 rooms (DISK: 3 of 6); the layout is unverified, so it is an option, not the default. A 15% overlap tolerance produced plans that looked broken; rooms now never overlap and unlinked rooms are drawn apart, marked "not connected".
- **Whole-house video** (E34): 17 COLMAP pieces, joins that rescale pieces 0.7–2.5×, walls duplicated and rotated 10–20°, rooms merged. Not usable for multi-room plans from this capture.
- **A damage query that names a surface** ("a cracked ceiling") boxed the clean ceiling. Negative queries for undamaged things fixed it (E24).

5. Fix loop (fixloop/)

Declared before any code change (commit 5baa29e, tag fixloop-before). The worst gate was video short side –35.8% and area –34.5%. The cause: only COLMAP's largest piece was used, 35% of frames. The prediction: within $\pm 10\%$ / $\pm 12\%$ after joining pieces. After (tag fixloop-after): **–4.0% short side, –8.1% area, interval now holds**. Predictions met, $\pm 3\%$ gate still failed as predicted. What remains: 36% of frames never register, bridges have $\sim 10^\circ$ rotation error through fast turns, and the capture is off-protocol.

6. Failure modes

Condition	Expected effect	Mitigation in place	Tested?
Mirrors	depth sees a "room" behind the glass. LiDAR/learned depth put points beyond the wall, so a see-through <i>opening</i> can appear on a mirror wall, and the room can leak	wall detection requires vertical coverage over 0.2–1.8 m, so a mirror's frame and the wall around it usually keep the wall; a see-through opening needs a run of ≥ 0.55 m with <i>no</i> wall surface	No. E1: the glass/mirror bathroom made DA3-Base fail pose estimation entirely (0/8), while Large managed 8/8
Glass (windows, glass doors)	LiDAR passes through, giving points outside	recorded as an opening: lowest see-through point > 0.5 m is typed <i>window</i> , otherwise door (≤ 1.2 m) or open passage	partly: bedroom window found (E15), width untaped
Wet-look / glossy floors	specular dropouts in LiDAR and DA3; reflections double the floor level	floor = lowest level with $\geq 40\%$ support + camera-height plausibility check, so a reflected floor below the real one has little support	No
Wet-look in damage	sheen read as a water stain	the prompt excludes reflections and dirt that wipes off; confidence ≥ 0.5 ; every region names its evidence frames for review	No (no damage captures yet)
Low light	fewer features (COLMAP pieces split), noisier DA3 depth, more ARKit drift	best-of-4 seeds; piece joining; drift correction. The per-piece metric-scale spread is recorded in plan.json but does not yet widen the intervals	No
Furniture against walls	outline edges snap to furniture faces	wall band starts at 0.2 m, so low furniture is excluded; tall furniture is not handled	Partly: E13 found outline edges snapping to furniture on sparse photo rooms (open issue)
Off-protocol capture (1× lens, HDR, mixed orientation)	narrower field of view breaks overlap; mixed shapes broke windows	<code>scripts/check_capture.py</code> audits a capture against the protocol; mixed orientations are letterboxed to one shape; the video tier warns on lens and HDR	Yes (E9: all original captures off-protocol, all warned)

The rule in every case: when evidence is missing, report nothing and say why rather than a confident wrong number. Examples: no ceiling when the floor is implausible, unlinked rooms drawn apart, empty damage without a detector.

7. Limits and next steps

- **Ground truth is thin:** one taped room. Tape for the other five rooms, door widths and second readings would allow interval calibration and opening-width scoring.
- **LiDAR accuracy and the magicplan head-to-head** need a Pro iPhone for an hour.
- **Damage recall** needs staged damage photographed with tape. The local detector needs no key (E24). Its 0 false positives on 17 clean photos is in-sample.
- **Video:** neither looser registration nor learned matches helped (E22, E23). Next would be DA3 poses throughout, with COLMAP only as a scale and loop anchor, plus a protocol emphasis on slow turns.
- **Photos:** rooms without a look-through photo stay unplaced. Door geometry alone was ambiguous on house_b (E25).

Benchmark report

Pipeline 0.8.0 on an M1 MacBook Air (8 GB, lite profile) unless noted. Every number below can be regenerated from the commands given; the experiment write-ups behind them are in `docs/EXPERIMENTS.md` (E-numbers).

What ground truth exists. Tape is available for one room only so far: the study (walls 312.4 / 330.2 / 312.4 / ~350.5 cm, ceiling 274.3 cm, single readings; `bench/ground_truth/home_tape.yaml`). The other five house rooms, door widths and second readings are still to be taped. The LiDAR captures are public Stray Scanner scans with no tape, so LiDAR is reported for self-consistency and drift, not accuracy. Claims are limited to what this supports.

1. Gates per tier (study, scored by `bench/gates.py`)

Gates: walls photo $\pm 8\%$, video $\pm 3\%$, LiDAR ± 2 cm or 1% ; area photo $\pm 10\%$, video $\pm 5\%$, LiDAR $\pm 2\%$; ceiling ≤ 1.5 cm; opening widths ≤ 2 cm on $\geq 85\%$. Walls are scored as the two side lengths against the mean of the opposite tape walls (long side 3.404 m: the two long walls differ by a cove).

Tier · capture	Short side	Long side	Floor area	Ceiling	Openings	90% intervals hold
Photos · house_b_study (protocol-style, 4 × 0.5× landscape)	3.074 m, -1.6% PASS	3.557 m, +4.5% PASS	10.71 m ² , +0.7% PASS	2.723 m, -2.0 cm FAIL	not taped	area yes, ceiling yes
Video · study_walk1.MOV (off-protocol 1×, after fix loop)	3.000 m, -4.0% FAIL	3.168 m, -6.9% FAIL	9.48 m ² , -10.8% FAIL	not reported	2 doors found, widths not taped	walls yes (since E21's 9.6% term), area yes
LiDAR	no taped LiDAR capture (needs a Pro iPhone)					

```
roomscan run data/raw/photos/house_b_study -o runs/bench && python bench/gates.py
runs/bench/house_b_study/plan.json
roomscan run data/raw/video/study_walk1.MOV -o runs/bench && python bench/gates.py
runs/bench/study_walk1.MOV/plan.json
```

Reading: the photo tier passes its wall and area gates on the one taped room and misses the ceiling gate by 0.5 cm (the interval still contains the tape). The ceiling error (-2.0 cm = -0.7%) is the same size and sign as the wall scale error (short side -1.6%): it is metric-scale noise from learned depth, not a ceiling-specific bias, so it is not tuned away on this single room (a correction fitted to one number would be fitting the test). The video tier fails: see the fix loop (`fixloop/RESULT.md`) for the declared worst gate and what remains (36% of frames never registered, DA3 bridges through fast turns, off-protocol capture).

2. Repeatability (live reruns, caches off)

`python bench/repeat.py <capture> --runs 3`: each run is a separate process with `ROOMSCAN_NO_CACHE=1`, so every model runs again. Results in `runs/repeat/<capture>/repeat.json` (committed copy: `bench/results/`).

Capture · tier	Runs	Short side	Long side	Floor area	Ceiling	Openings	Gates (each run)	Runtime
house_b_study · photos	3	3.0737 m, range	3.5567 m, range	10.7097 m ² , range	2.7229 m, range 0	1, 1, 1	walls ✓ area ✓ ceiling ✗	27.9 ± 1.1 s (cached)

Capture · tier	Runs	Short side	Long side	Floor area	Ceiling	Openings	Gates (each run)	Runtime
		0	0	0			identical	replay: 7.5 s)
study_walk1.MOV · video	3	2.9998 m, range 0	3.1683 m, range 0	9.4841 m ² , range 0	not reported	2, 2, 2	all $\times$ ($\pm 3\%$), identical	357 $\pm$ 34 s (cached: 112 s)

The photo tier is **bit-for-bit repeatable** on this machine: DA3 runs on MPS with a fixed input order and no sampling, and every downstream step is seeded. The runtime gap to a cached replay (28 s vs 7.5 s) confirms the models really ran. Repeatability across machines (CUDA vs MPS) is not measured; small float differences there are expected. The video tier is also identical across 3 live runs (COLMAP reran each time: 330–395 s vs 112 s cached). Before COLMAP was seeded and single-threaded, two runs of this video tracked 92 vs 65 frames (E10).

LiDAR is deterministic by construction (no learned model; seeded sampling): two runs of c7d28f72c6 give identical room areas and ceilings (checked in E12's regression run and again after E19).

2b. Interval calibration (all captures of the taped study, E26)

`python bench/calibration.py runs/e26/*/plan.json → bench/results/calibration_study.json.`

Quantity	Photo coverage (10 captures)	Video (1)
short side	8/10	1/1
long side	9/10	1/1
floor area	9/10	1/1
ceiling	2/2	n/a

Only one room is taped and several capture sets reuse photos, so this supports the error model; it is not a statistical calibration. The off-protocol scale term (15%) was fitted on these same captures.

3. LiDAR self-consistency and drift (no tape)

Capture	Rooms	Footprint	Ceilings	Adjacency	Drift (revisit misalignment)	Runtime
c7d28f72c6 (215 s walk)	5	63.52 m ²	2.27–3.07 m	4 links (3 doors, 1 passage)	19.7 → 3.5 cm with correction	212 s
c00a170fe1	3	21.38 m ²	not observed (floor-only scan)	3 links (1 probably false)	no revisit passed the test	25 s

Drift correction on/off (E7): wall sharpness 1.255 → 1.292; on the short walk no revisit is accepted and nothing moves, by design.

4. Connected plans

Tier	Capture	Rooms	Connected	How
Photos	house_b (17 photos, 6 rooms)	6	3 (living, kitchen, bathroom)	DISK+LightGlue links, two-way pose agreement, shared-wall snap (E17–E18)
LiDAR	c7d28f72c6	5	all 5	one cloud; shared doors + carved passages (E19)
Video	house_b_walk.MOV (212 s, 6 rooms)	3 (rooms merge)	all 3 via open passages	62% of frames tracked, 16/18 pieces joined (E21); not usable

The three unlinked photo rooms have no photo that sees into a neighbour; the protocol now asks for one look-through photo per doorway (E14).

5. Timing and memory (M1 Air 8 GB, lite profile, live = no cache)

Run	Runtime	Peak memory (phys_footprint, incl. GPU)
Photos, one room (4 photos), live	14.7 s	2.99 GB
Photos, full house (17 photos, 6 rooms, with stitching), live	135.7 s	4.22 GB (cross-room matching at the 4.5 GB GPU cap)
Photos, full house, replayed from checkpoints	10.1 s	0.44 GB
Video, study (64 s), COLMAP replayed, depth + join live	152.9 s	3.18 GB
LiDAR, 215 s walk with drift correction	311 s	0.35 GB

Memory is measured as macOS `phys_footprint` by `scripts/run_guarded.sh` (2 s sampling). Until 2026-10-03 the watchdog read RSS, which omits GPU (MPS), compressed and swapped memory: it reported ~0.65 GB for photo runs that really use 3–4 GB, and it let a 1698-frame video run thrash 2.1 GB of swap unnoticed. Earlier "peak" figures in EXPERIMENTS are RSS and understate memory. The full-house live runtime here (136 s) is longer than the 47.7 s in E17, which was measured with the cross-room links already checkpointed.

5b. Clean-machine test (README → result)

Fresh `git clone`, **empty** uv, Hugging Face and torch caches (everything downloaded: 1.6 GB packages, 1.7 GB weights), README steps timed on the M1 Air (data from the release zips via `ROOMSCAN_DATA_URL`, not yet uploaded):

Step	Time
<code>scripts/setup.sh</code> (+ web extra)	25 s
<code>scripts/setup_models.sh</code> (torch, DA3, pycolmap, kornia + weight prefetch)	320 s
<code>scripts/fetch_data.sh</code> photos lidar_single_room video_study (checksums verified)	3 s
<code>roomscan run LiDAR c00a170fe1</code>	59 s
<code>roomscan run photos house_b</code> (6 rooms, live models)	203 s
<code>pytest -q</code>	38 passed, 65 s
Total, clone to two plans	≈ 10 min

The first attempt failed and found two bugs: every run crashed without the `anthropic` package, and `setup_models.sh` did not install `kornia`. Both were fixed (b9e01b6) before this run. The fresh install reproduced the same room areas as the development environment (study 10.71 m², kitchen 6.22 m², ...). The photo run took 203 s here vs 48 s warm, because it shared the CPU with a COLMAP job and was the first model load.

5c. Robustness sweep (walk-in readiness)

`bench/local/sweep.sh` runs every capture on disk through the current pipeline. 2026-10-03: 22/25 completed at first; the 3 failures were two MPS out-of-memory crashes on fresh multi-room photo captures (a real walk-in risk, caused by the damage detector running between rooms; fixed) and one COLMAP child failure that did not reproduce. After the fix: **25/25 complete** (19 photo sets incl. mixed lens/orientation variants, 3 LiDAR scans, 3 videos), every plan schema-valid.

6. Damage, flags and scope

The chain runs end to end on all tiers (detector → projection onto floor / ceiling / wall → rules → scope), validated with a synthetic detector (schema-valid output; regions 0.45–1.1 m on the study walls). No accuracy number yet: it needs staged damage photographed with tape and an Anthropic API key for the detector.

Not measured yet (and why)

- Opening widths: door widths not taped.
- LiDAR accuracy and the magicplan head-to-head: need a Pro iPhone.
- Interval calibration (coverage $\approx 90\%$): one taped room gives 4–5 checks, too few to calibrate.

Figures

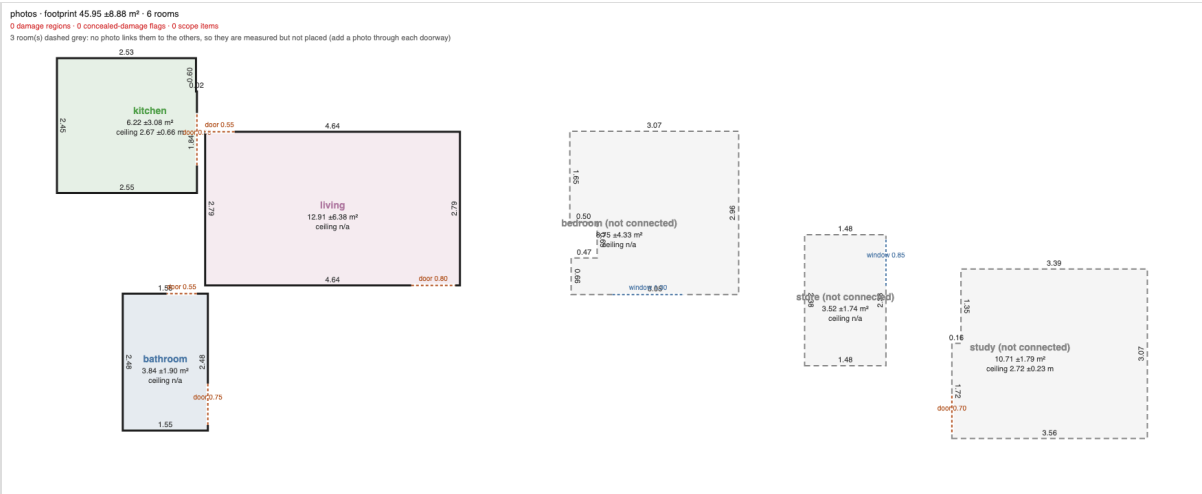


Photo tier, house_b (default matcher): 3 rooms linked; 3 measured but not linked, drawn dashed

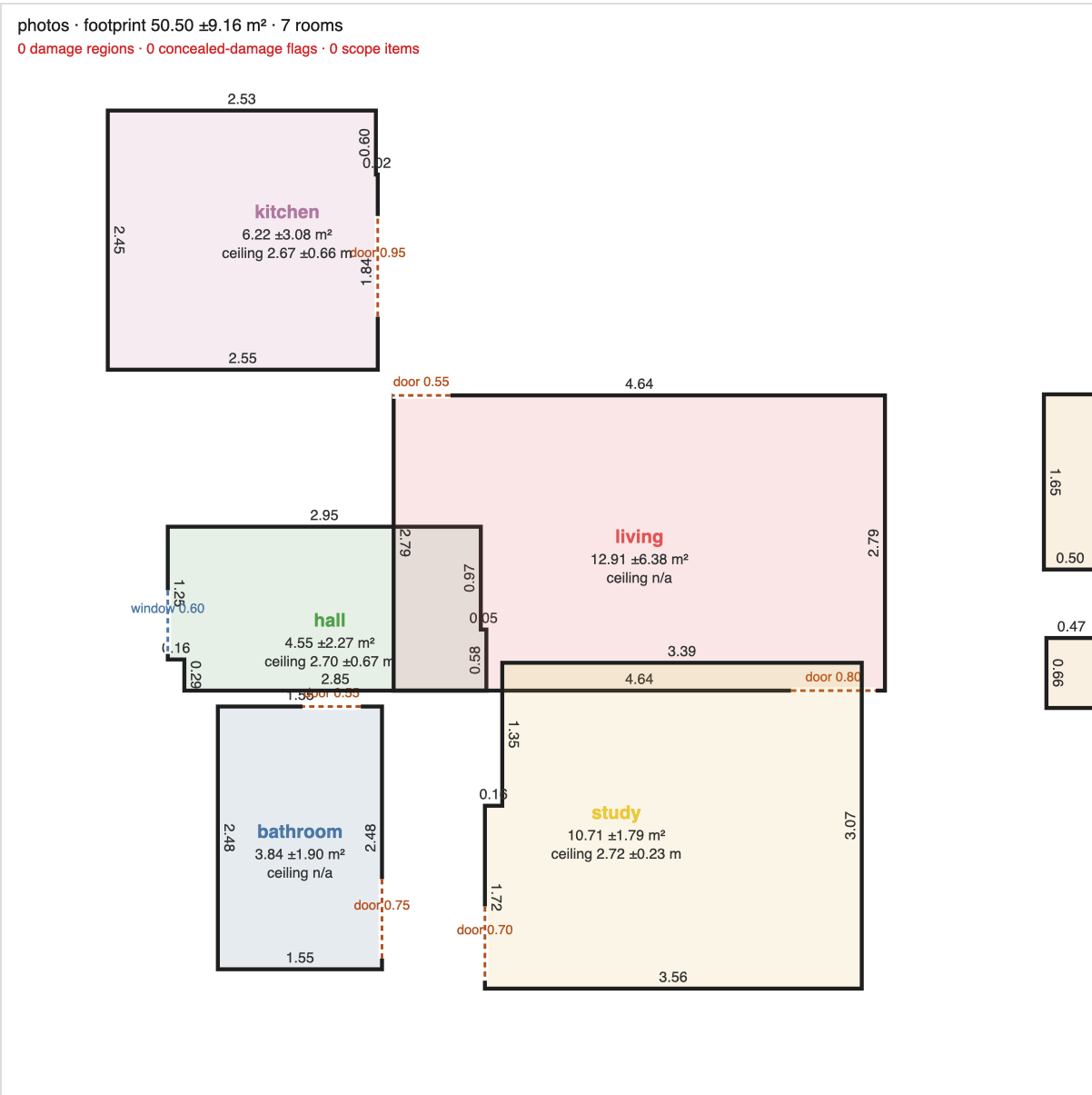


Photo tier with MAST3R matches on a GPU (E32): 5 of 7 rooms linked, layout unverified

house_b_walk.MOV: wall band (grey), keyframe cameras by COLMAP piece, plan rooms (black)



Why the whole-house video fails (E34): walls duplicated across 17 COLMAP pieces